Total Survey Error Framework

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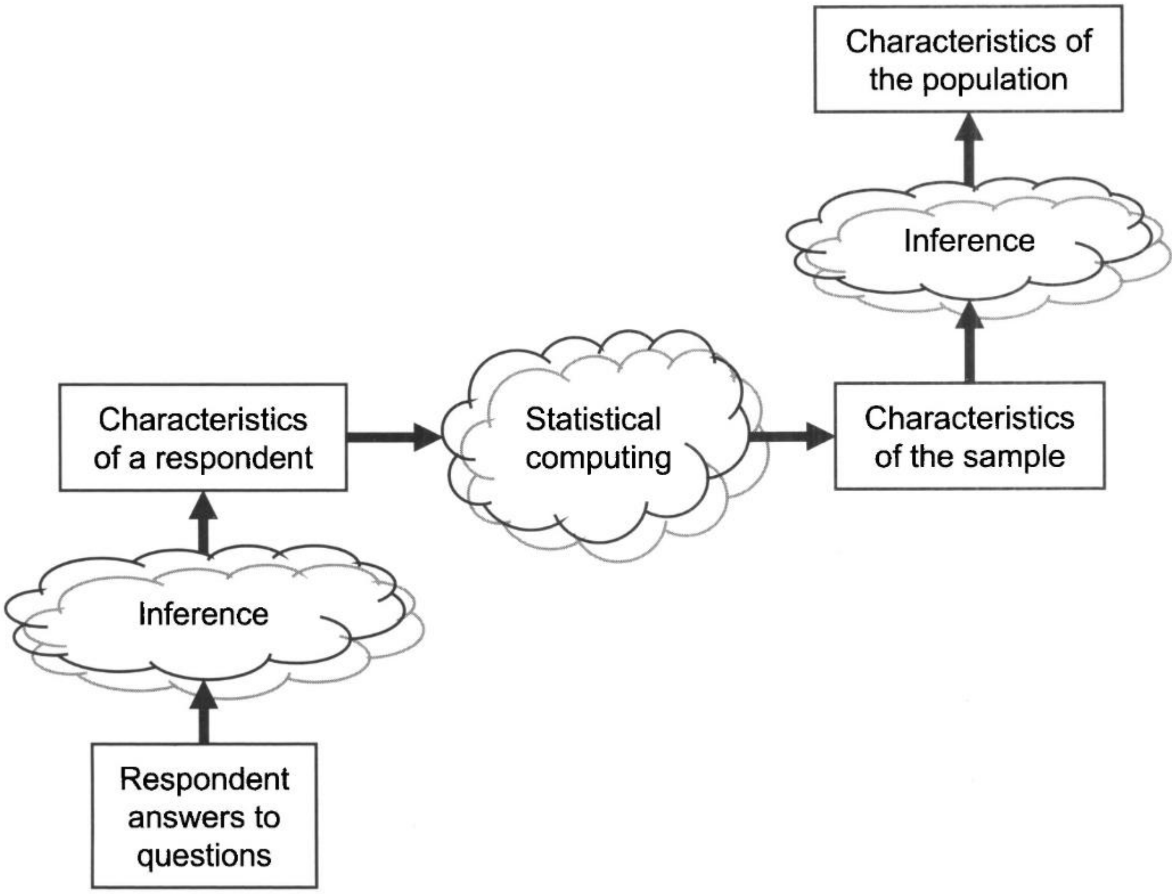
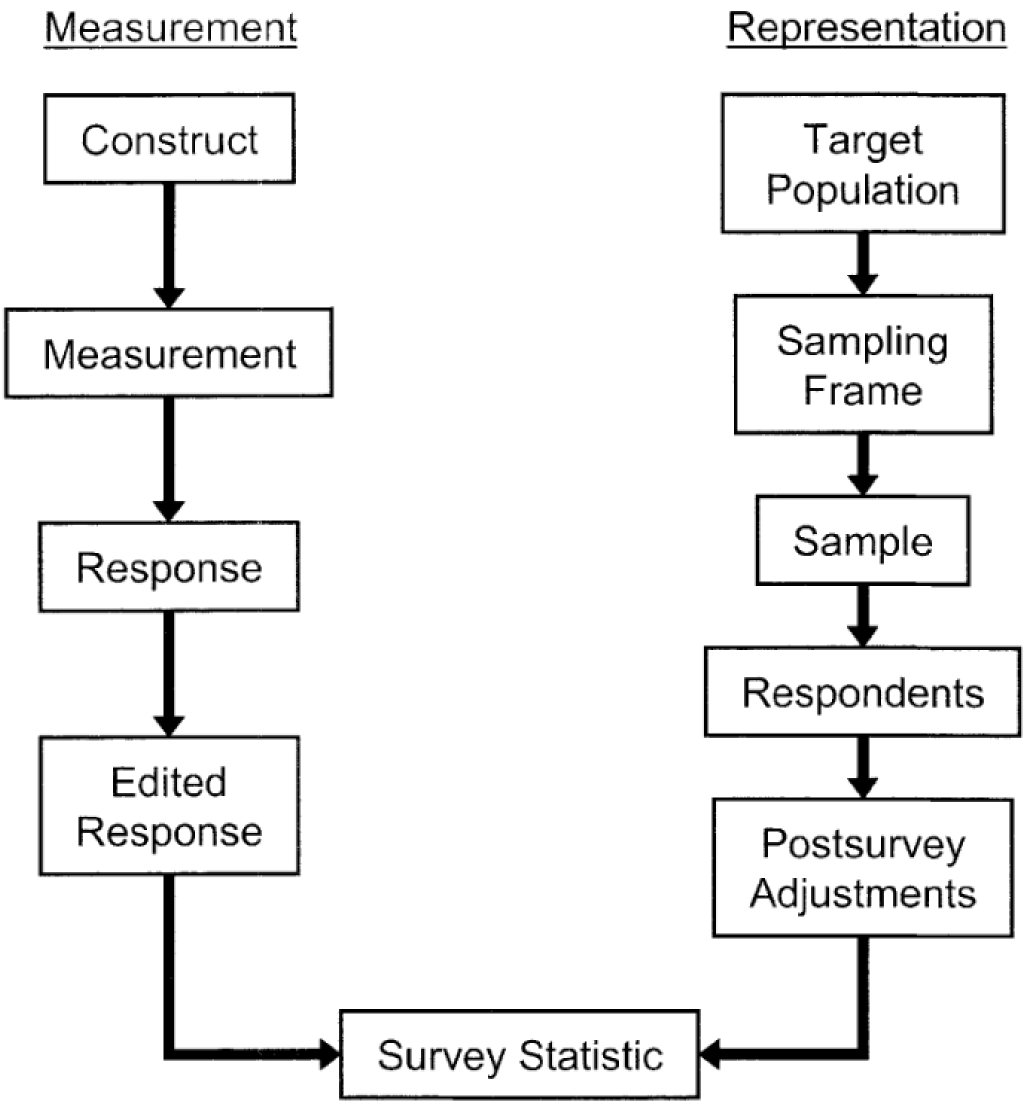
Inferences in Survey



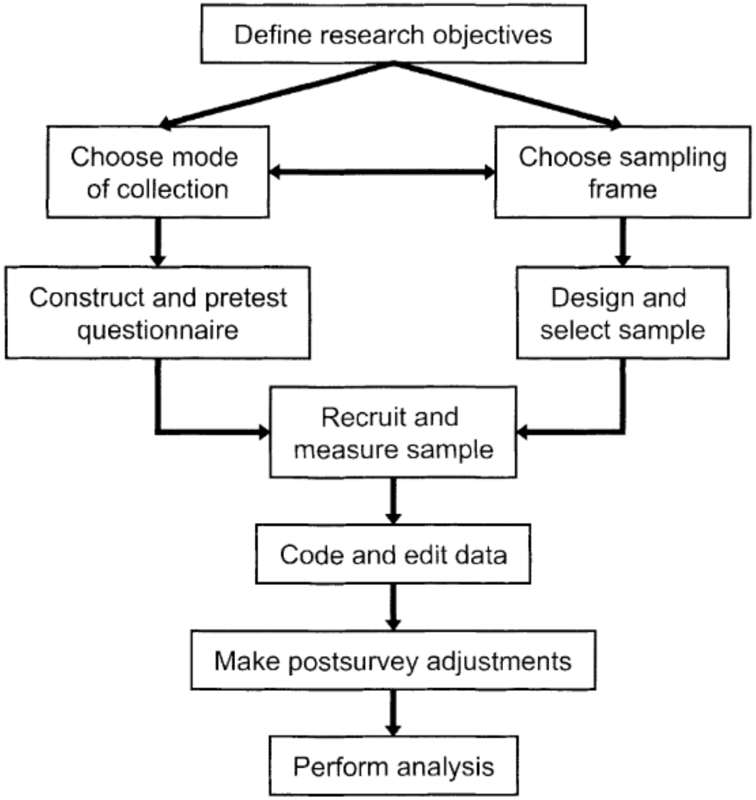
Inferences in Survey

- Answers people give must accurately describe characteristics of the respondents.
- The subset of persons participating in the survey must have characteristics similar to those of a larger population.
- When either of these two conditions is not met, the survey statistics are subject to "error."

Survey lifecycle from a design perspective

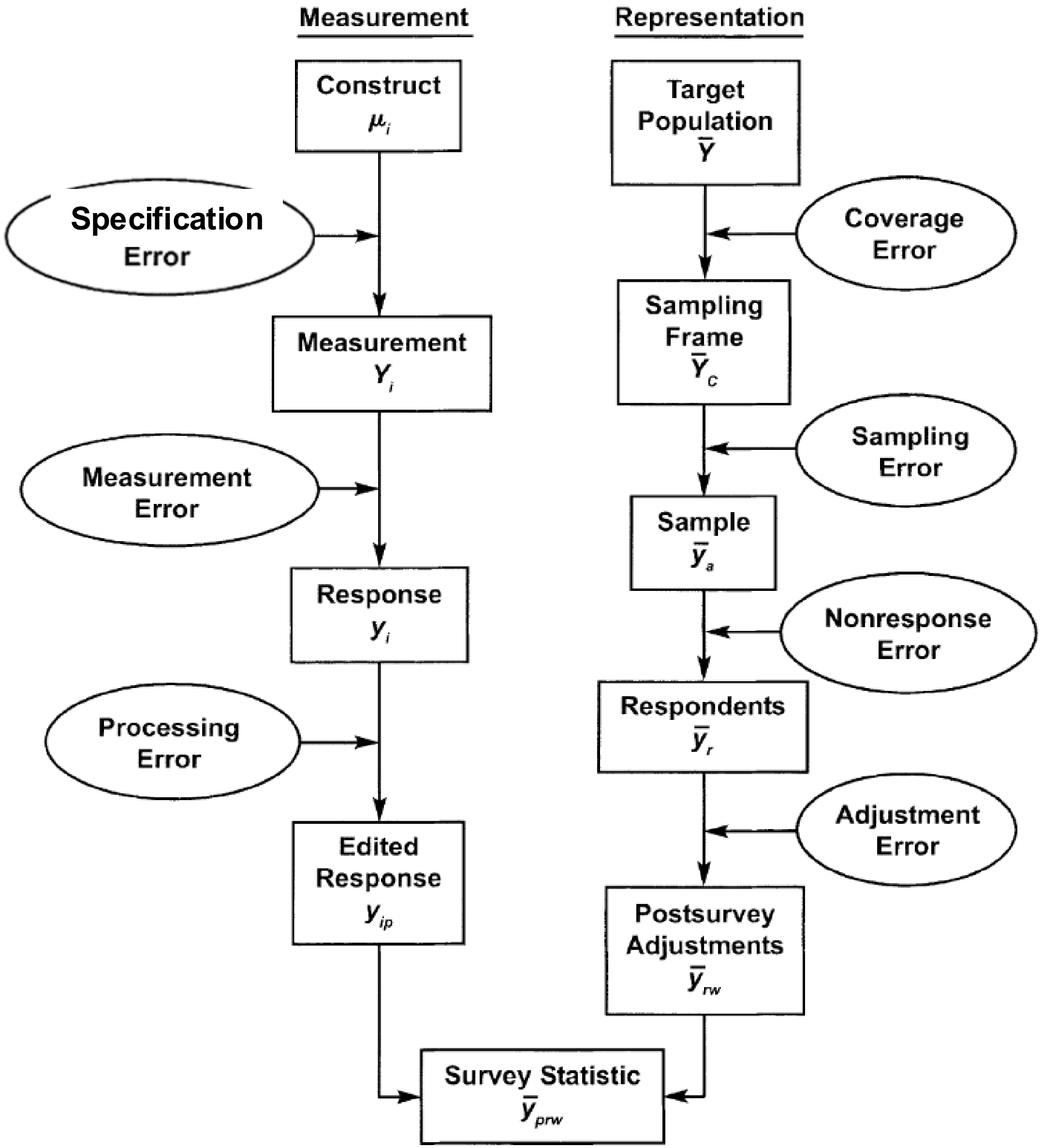


A survey from a process perspective



Survey lifecycle from a quality perspective

Survey error occurs when the estimate that is produced using survey data is different from the true value of the variable in the population one hopes to describe



- Total survey error (TSE) is the difference between a population mean, total, or other population parameter and the estimate of the parameter based on the sample survey (or census).
- TSE is the accumulation of all errors that may arise in the design, collection, processing, and analysis of survey data
- TSE concept was developed by Robert Groves (1989) in book on Survey Errors and Survey Costs

The Total Survey Error Framework

- There are several main types of error that surveyors must attempt to minimize
- No source of error can be ignored

The Total Survey Error Framework

- Surveyors often make the mistake of attempting to reduce one particular source of error while ignoring others
- Instead, surveyors should think in terms of minimizing total survey error (TSE)
- This involves attempting to maximize data accuracy by simultaneously controlling all four sources of error to the extent practical and possible, within the time, cost, and other constraints of the survey

TSE = sampling errors + non−sampling errors

- Sampling errors can be estimated and adjusted for probability-based sampling and are due to selecting a sample instead of the entire population
- Non-sampling errors are errors due to mistakes or system deficiencies, also from incomplete responses to the survey or its questions, etc.

Mean Squared Error (MSE)

- MSE is the sum of the total bias squared plus the variance components for all the various sources of error in the survey design.

$$\text{TSE MSE} = \text{bias}^2 + \text{variance}$$

- Systematic Error (bias)
    - errors that tend to agree
    - results in biased estimates
    - does not cancel out

Mean Squared Error (MSE)

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$$\text{TSE MSE} = \text{bias}^2 + \text{variance}$$

- Random Error (variance)
  - errors that tend to disagree
  - affects the variance (stability) of estimates
  - vary from case to case but are expected to cancel out

Mean Squared Error (MSE)

- MSE is a metric for measuring TSE
- MSE cannot be calculated directly but useful conceptually to consider how large the different components can be and how much they add to the total survey error
- Hypothetical but great guide for optimal survey designs
- Survey design goal is to minimize the MSE
- When other designs are similar on other quality dimensions, the optimal design is the one achieving the smallest mean squared error

Coverage Error

- Occurs when the list of members of the target population from which sample members are drawn (the *sampling frame*) does not accurately reflect the population on the characteristic(s) the survey is trying to estimate
- Coverage error is the difference between the estimate produced when the list is inaccurate and what would have been produced with an accurate list
- A high-quality sample survey requires that every member of the population has a known, nonzero probability of being sampled.
- The ideal case is when there is a one-to-one correspondence between the frame and target population units (i.e., each target unit is included once and only once on the frame, and the frame does not contain units that are not in the target population)

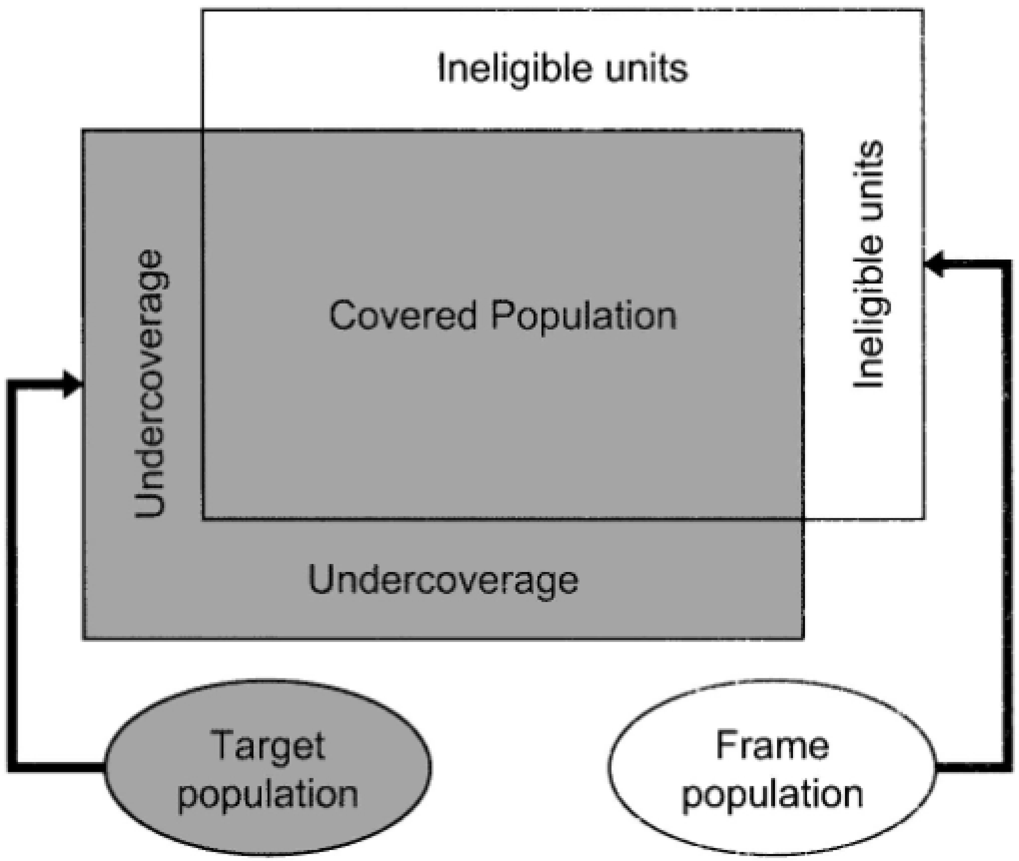
Undercoverage

- If some units in the population are not on the list of the frame, there will be *undercoverage*
- Those not on the list will have zero probability of being selected
- When under-coverage occurs, the survey researcher faces three options:
  - Revise the sampling frame or use multiple frames,
  - Redefine the target population to fit the frame better,
  - Admit the possibility of coverage error in statistics describing the original target population.

Overcoverage

- Overcoverage occurs when
  - Some units in the frame are not members of the target population (i.e., they are *ineligible* or *out of scope* for the survey) or
  - Some units have higher chance of being selected due to duplicated IDs
- If cannot be removed a priori, the ineligible units are usually identified during the data collection and deleted with no biasing effects on the survey data.
- Some units that are out of scope for the survey are mistaken for eligible units, and consequently, the estimates based on the survey results may be biased

Coverage Error



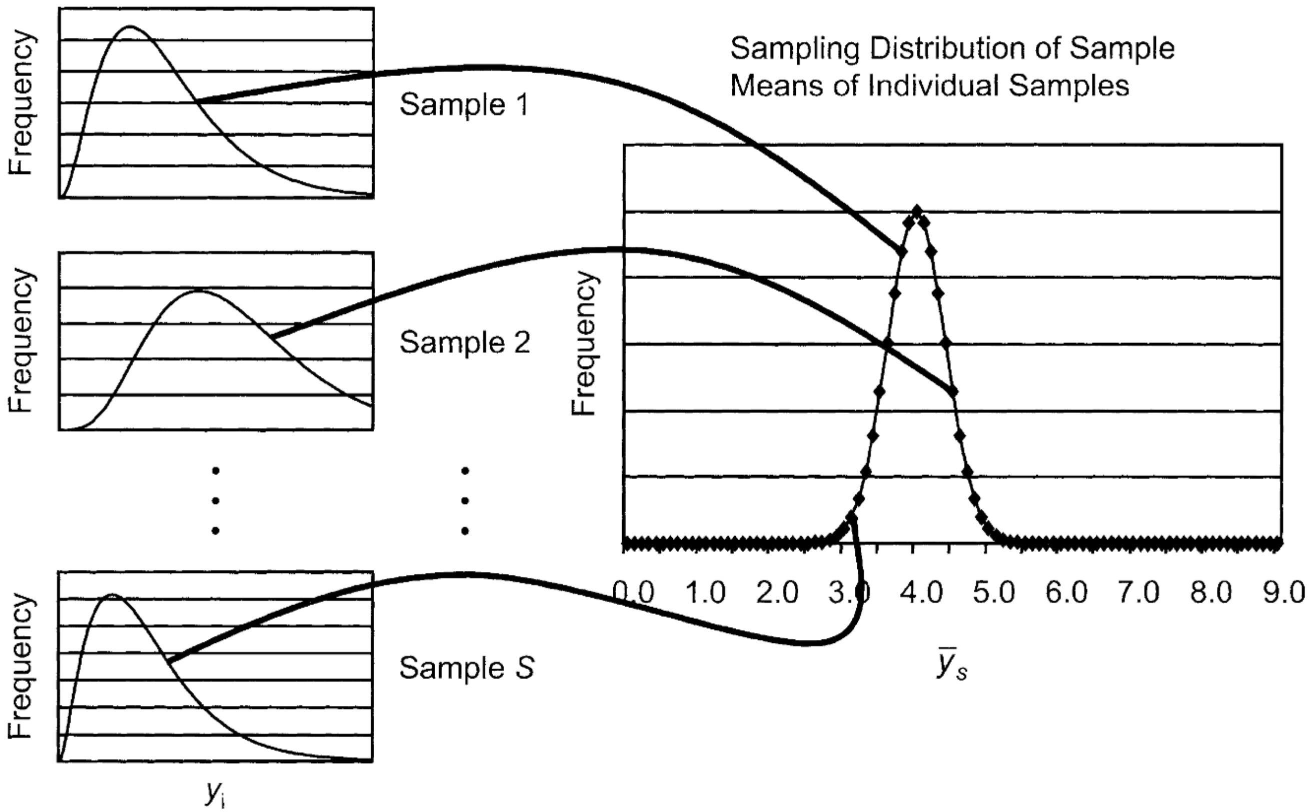
Sampling Error

- The difference between the estimate produced when only a sample of units is surveyed and the estimate produced when all units are surveyed
- Occurs any time a survey includes only some, rather than all, members of the sampling frame; it is *unavoidable* in sample surveys
- But it can be minimized, and probability sampling allows us to obtain estimates about the population within acceptable levels of precision by surveying only a small portion of the population

Sampling Error

- There are two types of sampling error: sampling bias and sampling variance.
- Sampling bias arises when some members of the sampling frame are given no chance (or reduced chance) of selection.
- Sampling variance arises because, given the design for the sample, by chance many different sets of frame elements could be drawn

Sampling Error



Sampling Error

- The extent of the error due to sampling is a function of four basic principles of the design:
  - Whether all sampling frame elements have known, nonzero chances of selection into the sample (called "probability sampling")
  - Whether the sample is designed to control the representation of key subpopulations in the sample (called "stratification")
  - Whether individual elements are drawn directly and independently or in groups (called "element" or "cluster" samples)
  - How large a sample of elements is selected

Sampling Error

- Sampling bias can be easily removed by giving all elements an equal chance of selection.
- Sampling variance is reduced with
  – big samples,
  – with samples that are stratified, and
  – samples that are not clustered.

Nonresponse Error

- The difference between the estimate produced when only some of the sampled units respond to the survey compared to when all of them respond
- It occurs if those who don’t respond are different from those who do in a way that influences the estimate
- A high response rate can reduce the likelihood of nonresponse error, but it does not necessarily mean that there is low nonresponse error

Nonresponse Error

- Unit nonresponse occurs when a sample unit does not response to any part of the survey
- Item nonresponse occurs when the survey is only partially completed, and some items are not answered
- Incomplete response occurs when the response to open-ended question is incomplete or very short and inadequate
- Panel attrition occurs when a sample unit is lost over the period of a longitudinal study

Nonresponse Error

- Nonresponse bias for the sample mean is the product of the nonresponse rate (the proportion of eligible sample elements for which data are not collected) and the difference between the respondent and nonrespondent mean.
- This indicates that response rates alone are not quality indicators
- High response rate surveys can have high nonresponse bias
- High response rates reduce the risk of nonresponse bias

Measurement Error

- The difference between the estimate produced and the true value because respondents gave inaccurate answers

$$X = T + E$$

Measurement Error

- Respondents may be unwilling or unable to provide accurate answers for a variety of reasons: poor question design, survey mode effects, interviewer and respondent behavior, or data collection mistakes
- even if the question is a good way to measure a specific concept, there are many other ways in which measurement error can still occur

Measurement Error

Random error

- Random in nature
- Will increase and decrease a person's score by exactly the same amount with infinite testing
- Cancels itself out
- Lowers reliability of a test

Systematic error

- Occurs when source of error always increase or decrease a true score
- Does not lower reliability of a test since the test is reliably inaccurate by the same amount each time

Measurement Error

- Standard error of measurement (SEM) is defined as the standard deviation of an individual's observed scores around their true score
- SEM is an index of the amount of inconsistency or error expected in an individual's observed test score
- SEM is an estimate of how much the individual's observed test score might differ from the individual's true test score
- SEM can be used to quantify the amount of *random* measurement error

Measurement Error

- Based on sample data, SEM can be calculated as

$$SEM = s_x\sqrt{1 - \rho_x}$$

where s_x is the standard deviation of observed scores and ρ_x is estimated reliability.

Measurement Error

- As an example, if the standard deviation for a sample of observed assessment scores was 15, and the reliability of the scale, as estimated using the coefficient alpha method was 0.75, then the SEM would be calculated as _________
- After obtaining the SEM, a 95% confidence interval can be constructed for observed score x +/- 1.96SEM
- The 95% confidence interval for an observed score of 87 is [_________,________]

Specification Error

- Specification error may occur when there is a mismatch between what the survey is measuring and what it is intended to measure
- When psychological tests are used, this is about the validity of the test scores

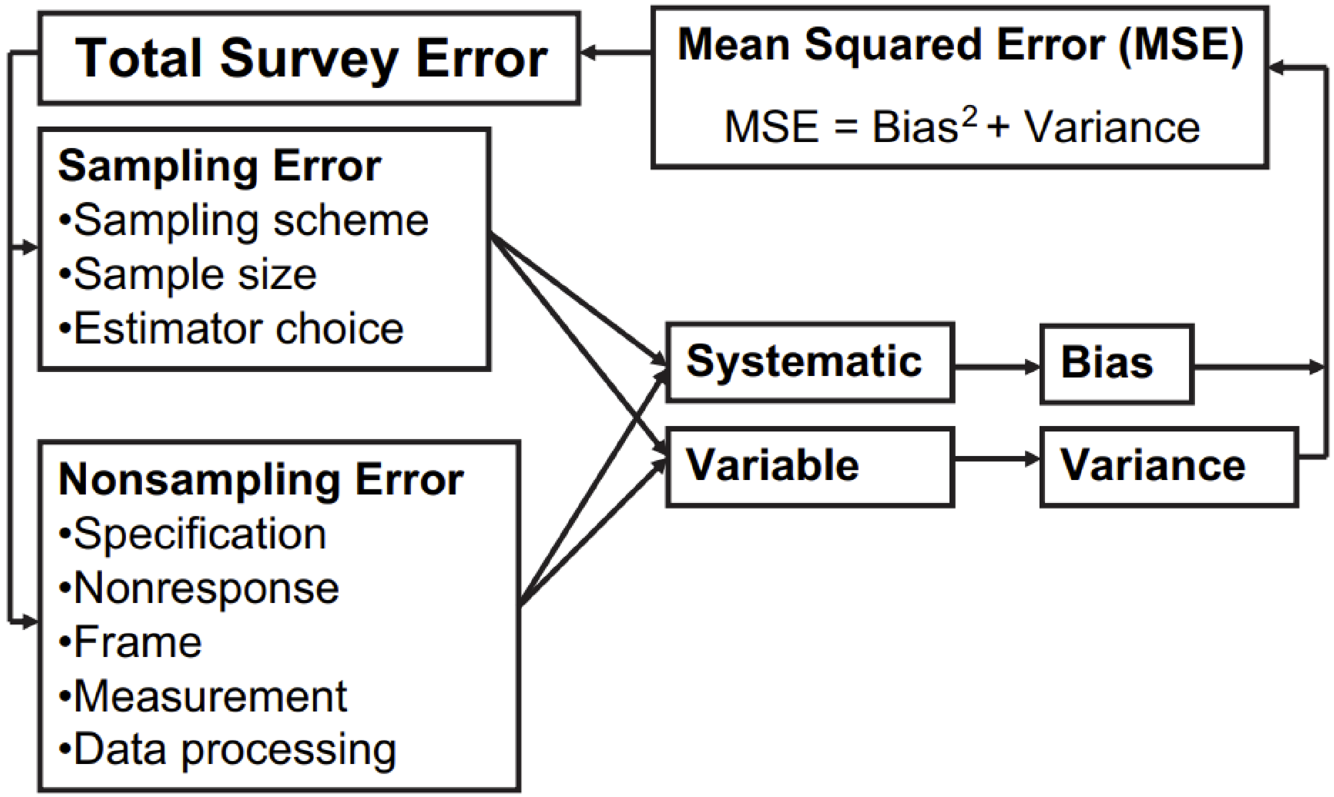
Specification Error

- Bureau of Labor Statistics (BLS) defines people “on layoff” as those who are separated from a job and **await a recall** to return to that job.
- ‘‘Were you on layoff from a job?’’ (a previous version before 1994)
- To many people, the term ‘‘layoff’’ could mean permanent termination
- ‘‘Has your employer given you a date to return to work?’’ and ‘‘Could you have returned to work if you had been recalled?’’ (new questions after 1994)

Processing Errors

- errors that arise during the data processing stage, including errors in the editing of data, data entry, coding, the assignment of survey weights, and the tabulation of survey data.
- For open-ended items that are coded, coding error is another type of data processing error.

Summary



Biemer (2010)